

NETWORK TRAFFIC REPORTING

TEAM MEMBERS: Pranav Chinthakuntla, Maggie Gao, Yatharth Jain, Manas Jha, Lakshay Mahajan, Adrian Maliackel, Riley Noggle, Derek Peng, Carlos Reyes, Nikita Sirandasu, Ashwin Tarinikanti



OVERVIEW

About Caterpillar:

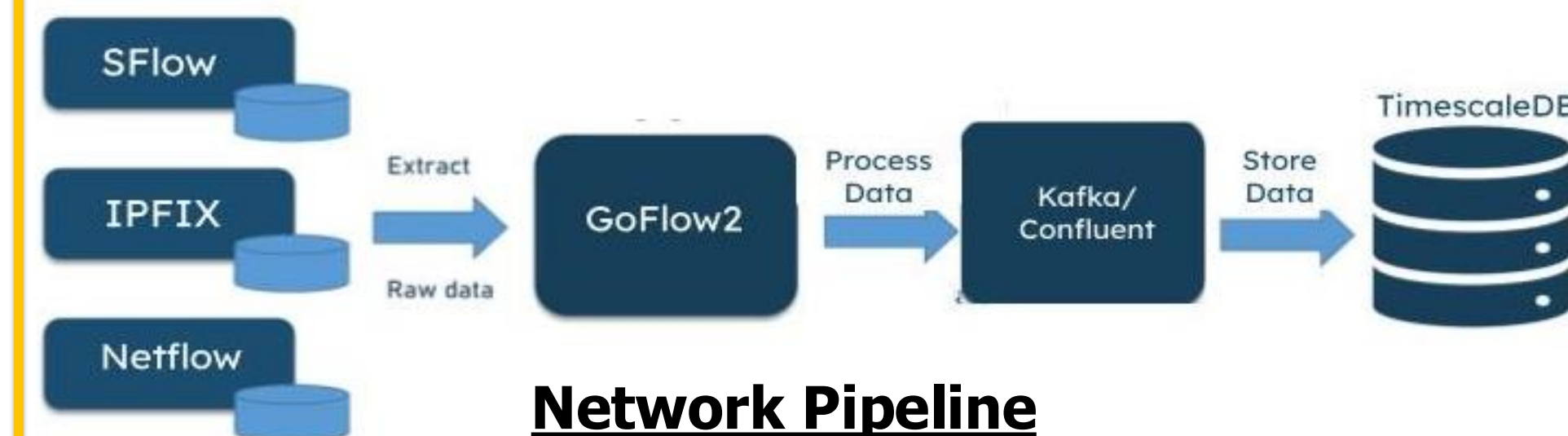
- Global industry leader in construction and mining equipment, off-highway engines, turbines, & locomotives

Project Objective:

- Our goal is to create a user-friendly and accessible enterprise traffic reporting system to replace Cat's legacy system. This system has many benefits, such as:
 - Providing **insights** and **global data analysis**, which will help efficiently **identify trends** and **anomalies**.
 - Helping support the IT department by **improving efficiency** and **effectiveness** of network monitoring and engagement.

PLATFORM TEAM

- Objective:** Develop a **pipeline** system for collecting, processing, and storing network traffic data with:
 - GoFlow2 – network traffic data **collection**
 - Kafka/Confluent - **processing and batching**
 - Kafka – buffer for data streams
 - Confluent – uses JDBC Connector to integrate Kafka to database
 - TimescaleDB – PostgreSQL database for **storage**
 - Set up logging system and developed PostgreSQL database schema for maintenance.

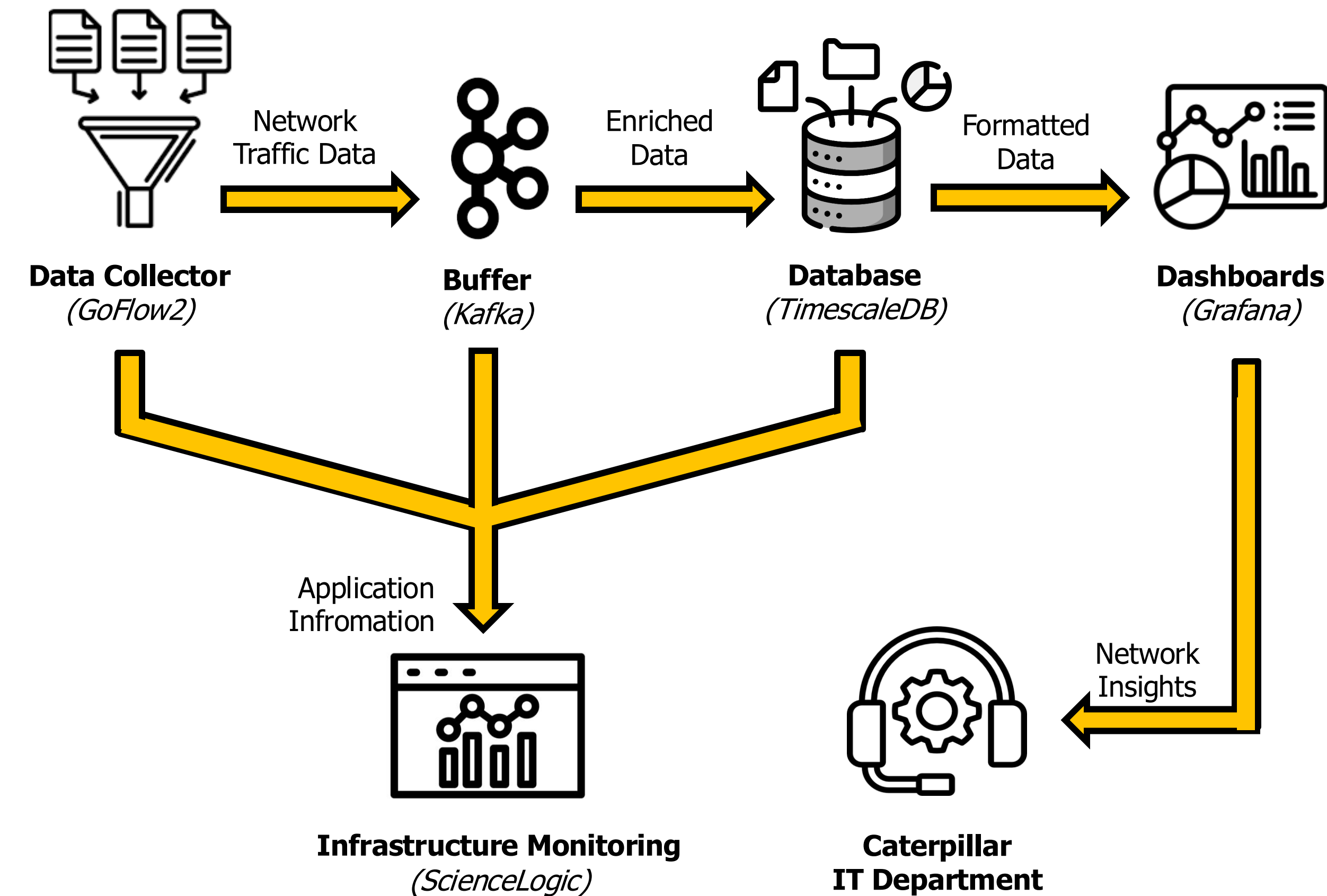


REFERENCES

- [Apache Kafka Documentation](#)
- [GoFlow2 Documentation](#)
- [How the OSI Model Works | Network Fundamentals](#)
- [TimescaleDB Documentation](#)

METHODOLOGY

Network Traffic and Application Insights



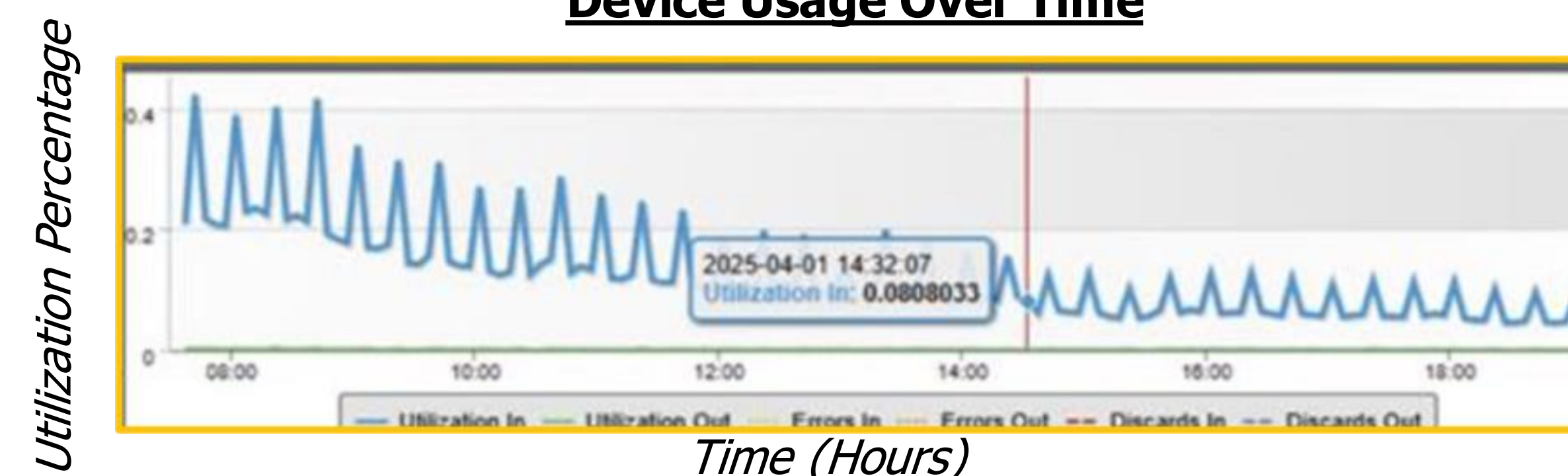
ANALYTICS TEAM

Objective:

 Monitoring, cleaning, and analyzing

- Investigated multiple methodologies across our various databases to extract **variable insights**.
- Used **ScienceLogic**—a monitoring platform—to integrate our database and monitor database processes in real-time.
- Created Python scripts to **analyze and clean ScienceLogic data** for improved database analysis.

Device Usage Over Time



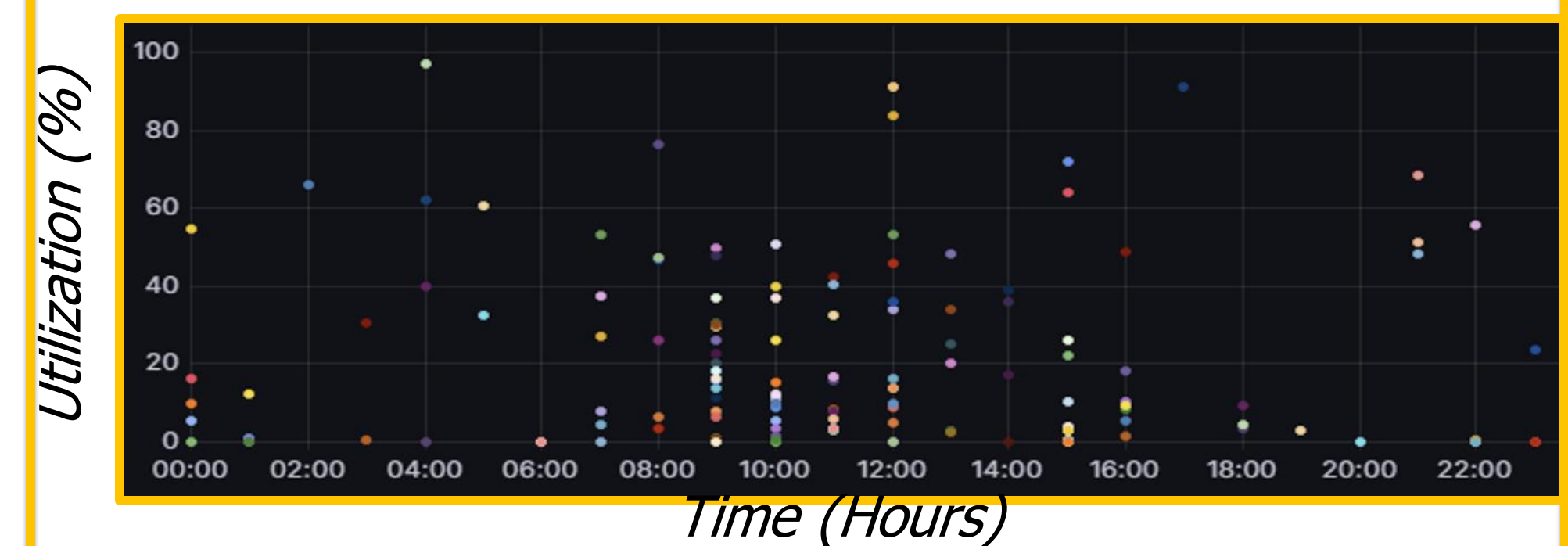
USER INTERFACE TEAM

Objective:

 User-friendly, intuitive, accessible interfaces

- Dashboards made with **Grafana**—a data visualization platform—with functions such as:
 - Displaying **network activity trends** during business and non-business hours globally.
 - Monitoring **device** and **process memory** usage.
 - Displaying **peak utilization patterns** during the day.

Peak Utilization Points Dashboard



CONCLUSION & FUTURE PLANS

- Platform Team:** Built a **network data pipeline** and added **database monitoring** for efficiency.
- Analytics Team:** Developed **visualizations** and Python **scripts** for ScienceLogic data analysis.
- UI Team:** Created **dashboards** and improved network traffic queries.
- Future Plans:** These efforts have strengthened CAT's data infrastructure, **improving performance** and **analytical capabilities**. CAT plans to use our system as a supplemental tool to **diagnose** and **troubleshoot issues** and **enhance logging analytics** next year.

ACKNOWLEDGEMENTS

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